

Raised-energy CFF stack

Functionality · Test contracts · Rust idioms · KISS

2026-09-14 · Seven functional commits and documentation

Assessment

The stack contains seven functional commits and one documentation commit. It combines exact generalized CFF generation, local and integrated UV reconstruction, physical phase conventions and command workflows. The additional optimization commit owns canonical UV algebra and reuse of immutable graph preparation. All eight pinned commits use Lucien Huber <im@lcnbr.ch>.

Fresh validation: COMPLETE — 93 PASS of 93 planned stages. This report distinguishes static review, accepted test executions and measurements on explicitly recorded sources. Passing strategy comparisons support correctness; independent signed contours and complete reconstruction certificates provide mathematical oracles. Detailed commands, attempted executions and exclusions are in the [validation ledger](#).

Functional ownership

Commit	Pinned identity	Responsibility
C1	665658b16898	Symbolic/tensor foundations and graph bookkeeping
C2	a9397ee0f86b	Exact shared generalized CFF generation
C3	3ea313789a1a	GammaLoop adapters, local UV reconstruction and evaluator preparation
C4	04637884f24f	Command, persistence, evaluation and benchmark workflows
C5	1f2cf6d8236d	Physical phases and model sewing
C6	8e4e4664f3f9	D-dimensional integrated UV algebra and analytic scalar-product protection
C7	6d7a9220129c	Canonical UV algebra, bounded occurrence allocation and graph-owned reuse
C8	oxsmszsrort	Current reports, source attribution and validation evidence

Commit 7, 6d7a9220129c0fac37bdc0e04ebefccb1688c7be, pins the executable implementation. Commit 8 is identified by its stable jj change, avoiding a reference to its own final hash. The [source mapping](#) records full revisions and trees. The final bookmark and documentation delta are verified separately during the final report update and closure.

1. Symbolic and tensor foundations

Tensor parsing preserves scalar factors, open slots and independent dummy indices. Supported compact vectors accept scalar weights; ambiguous vector products inside scalar functions remain complete opaque arguments. Scalar aliases resolve through the existing result boundary. Odd powers repeatedly contract one fixed base square with the remaining tensor, preserving the requested exponent across all five leaf forms.

Graph rewiring retains slot order, edge flow and complete node/forest membership. Completed execution waves release unused tensor payloads while retaining live aliases and handles. Tests compare full contractions with independent component sums across dense, sparse and mixed storage, metric signatures and execution strategies. Factorization tests preserve scalar spectators. No universal contraction ordering or isolated speedup follows from these contracts.

2. Exact generalized CFF generation

The shared engine stores coefficients as `Rational` and treats repeated propagator occurrences independently, including their numerator capacities. Parsing handles arithmetic precedence, signs and checked exponents; source validation checks IDs, endpoints, dimensions and cut aliases before indexing. Zero and infinite-denominator inputs have explicit value contracts.

Complete signed residues cover simple/repeated poles, disconnected components, reversed routing and execution strategies. Reuse keys retain the relevant topology, capacities and contour state. Rational storage expresses the coefficient domain; it is not itself a measured performance claim.

3. Source adapters and evaluator preparation

GammaLoop owns physical incidence, denominator occurrences, energy assignments and residue-map construction. Immutable assignments bind capacity generation to the actual numerator mapping. Denominator momentum, mass, multiplicity and domain are certified separately from the complete signed numerator. A loop-momentum basis supplies coordinates, not ownership.

GL00/GL04 child certificates compare complete factorized numerators with independent Taylor oracles in a neutral chart; nested/disjoint tests cover surrounding forest composition. Finite tensor-sum boundaries may close compatible leaves before the normal contraction schedule, preserving separate sums and scalar spectators. Evaluator preparation stays at its existing numerical execution boundary and does not distribute graph numerators.

4. Commands and evaluation workflows

One placeholder parser defines command templates and lexical scope. Validation catches references and cycles; temporary settings and active blocks are restored after returned errors. Manifest-based saved-state handling keeps persistence behind explicit save operations. Unsupported versions fail before payload decoding or replacement, and CLI errors produce failing exits.

A Serde pair-list adapter preserves structured additional-weight keys through inspection output. Tests execute commands, reload exported functions and saved state, check complete weights and history, and verify restoration after success/failure. Timing fields distinguish expression construction, tensor preprocessing, orchestration and literal Symbolica evaluator builds; inclusive intervals are not interchangeable.

5. Physical phases and model sewing

Normalization retains complete propagator/vertex factors and applies the shared conversion at the cut-group boundary. Connected right-hand components include cut hairs; conjugate couplings and spin matrices enter once. Fixed-momentum bridge vertices remain unless their endpoints touch the initial cut.

Model multiplets validate spin, statistics, charge, color, mass, propagators and conjugation. Independent polarization sums, Ward identities, left/right mirrors and signed LO/NLO acceptances constrain complementary parts of the convention. Exact numerator matching precedes sampled fallback. Sampling remains probabilistic; arbitrary non-Hermitian interactions and complex-mass cutting rules are outside the supported contract.

6. D-dimensional integrated UV algebra

Projected-tensor and complete-numerator Vakint modes share Lorentz validation, coefficient restoration and Laurent-depth accounting. Coefficient/normalization poles determine the necessary master-integral depth. D-dimensional algebra closes before scalar dimension substitution, preserving evanescent finite contributions.

Collision-safe scalar aliases protect completed dot products during analytic spin expansion while exposing open tensor slots. Powered-dot freshening also remains necessary at the whole-numerator FORM boundary: the independent average of $(k \cdot p + k \cdot q)^2$ is $k^2 \cdot (p^2 + 2 \cdot p \cdot q + q^2) / D$, with $D = 4 - 2 \cdot \epsilon$. Forest composition retains physical mass owners: contained owners participate in Taylor expansion, disjoint owners remain fixed and partial overlaps are rejected.

7. Canonical local UV projection and reuse

Raw sectors retain physical owners and provenance for subsequent Taylor operations. A separate canonical projection view groups exact signed routing, mass, denominator polynomial/prescription and component domain.

Physical incidence remains an independent witness. Only certified serial paths or pure cycles permit equivalent channels to merge into powered quotients; nonadjacent channels retain their topology.

Completed hard factors may use an unambiguous certified class. Physical-source-fixed/soft factors keep their restrictions; hard factors without a surviving class remain affine carriers. Odd numerator signs are independent of denominator evenness. Complete numerator and denominator reconstruction certificates precede native CFF generation.

The compressed allocator admits a baseline, an eligible packed alternative and one placement challenger. Native source-map rows score the bounded proposals; rank and stable order break ties. This is not a global optimum. One nonserialized graph-owned context reuses immutable preparations, assignment templates and mapped subtrees under complete keys and bounded retention. Component waves and bulk cut-key addition combine complete coefficient values while preserving factorization.

Same-frame +N/ -N sectors may cancel during grouping. The grouped result returns the existing typed zero before requiring production maps, preserving every supported final cut order. The full projection/build regression tests that value rather than sector layout.

Behavioral contracts and findings

Small scalar identities use exact subtraction and zero checks. Factorization-specific tests compare the complete retained factorized expression. Tensor oracles permit dummy-index renaming while preserving physical labels. Cache/build counts, storage layout, traversal/proposal order and private row-count goldens do not define numerical correctness; public retention bounds and supported value/error behavior remain valid contracts.

Finding	Current contract	Fresh execution evidence
Projection cancellation	Complete typed zero at every supported final cut order	PASS: C7/uv-performance-outward, C8/curated
Private layout/cache assertions	Full values, supported errors, factorization and public bounds	Exact reviewed correction fingerprints in the source mapping; owning selections below
Physical route tolerance	Fixed 1e-9 complex-relative acceptance for GL00/GL01	PASS: C7/physical-local-uv-routes, C8/physical-local-uv-routes
Sunset allocation	Independent complete degree-five/seven signed contour values	PASS: C7/uv-performance-outward, C8/curated

The signed sunset oracle covers all 126 distributions of degree five over five repeated occurrences and admitted degree-five/seven plans. Independent successive clockwise contours at energies (2, 3, 7) give 1213/16387080192 and -365/47775744. The retained mathematical audit is separate from fresh application-test execution. GL00/GL01 route comparisons require finite, nonzero complex results at three fixed points within 1e-9 relative complex-norm tolerance. Reported stability estimates cannot relax that threshold; integrated and threshold counterterms are disabled to isolate this comparison.

Rust patterns and KISS

Pattern	Effectiveness and limit
Borrowed AtomView, owned Atom, concrete Rational	Avoid copying during inspection, retain immutable witnesses and enforce the coefficient domain. Domain clarity does not prove speed.
Typed references and domain enums	Separate owners, canonical classes, occurrences, residue maps and valid input modes; raw/canonical UV types protect the recursion/projection boundary.
Immutable Arc payloads in a graph-owned context	Keep assignments and generated payloads together; scoped lifetime permits reuse without serializing caches into state.
Existing builders and fallible conversions	Keep validation and numerical preparation at established owners. Validate complete inputs before mutation or dispatch.
BTreeMap::entry and consuming iterators	Accumulate exact values and release consumed state. Deterministic order aids diagnostics without becoming an oracle.
Fixed points and bounded proposals	Structural convergence or decreasing state counts support termination; proposal limits do not bound total runtime.

Pattern	Effectiveness and limit
Result accessors and Cow	Unify literal/lazy scalar interpretation and borrow when possible; unsupported interpretation returns an explicit error.
Temporary scalar aliases	Preserve completed scalar contractions during tensor expansion; collision checks prevent capture.

Shared preparation, immutable bindings, one scalar-result boundary and one bulk accumulation path remove duplication. Canonical source reconstruction and tensor-boundary rewiring remain substantial algorithms with explicit invariants. Further cache layers and diagnostic state need representative evidence to justify their complexity. Strategy comparisons do not establish universal optimality.

Fresh validation

Boundary	Format / check / build / Clippy	Owning behavioral selections
C1	Four gates PASS	3/3 selections PASS; 811 accepted executions
C2	Four gates PASS	3/3 selections PASS; 182 accepted executions
C3	Four gates PASS	1/1 selections PASS; 275 accepted executions
C4	Four gates PASS	2/2 selections PASS; 623 accepted executions
C5	Four gates PASS	4/4 selections PASS; 217 accepted executions
C6	Four gates PASS	1/1 selections PASS; 242 accepted executions
C7	Four gates PASS	2/2 selections PASS; 291 accepted executions
C8	Four gates PASS	10/10 selections PASS; 2706 accepted executions

Final selection	Accepted passed / executed	Status
Curated GammaLoop (includes phase/Vakint subsets)	2135 / 2135	PASS
Full scalar matrix, including slow cases	167 / 167	PASS
Vertex rules	1 / 1	PASS
UFO model parity (API library)	1 / 1	PASS
Physical local UV routes: GL00/GL01	2 / 2	PASS
Spenso: shadowing	168 / 168	PASS
Spenso: no default features	50 / 50	PASS
Shared CFF: default features	37 / 37	PASS
Shared CFF: all features	108 / 108	PASS
Shared CFF: no default features	37 / 37	PASS

Accepted final runtime: **2706 executions over 2279 distinct binary/test identities**, from 10/10 passing selections. Accepted configurations report 1572 excluded entries, which may overlap. Baseline and owning-prefix executions are excluded from these final totals. Phase/Vakint cases inside the curated selection are not added again as independent suites. Configuration overlap and owning-prefix executions remain distinct in the ledger. Pending, running, failed and unaudited stages prevent a complete result; zero accepted failures does not establish overall success.

The recorded configuration uses `--locked` and the `test_gammaloop nextest` profile. Workspace `check/build/Clippy` gates and most suites use the `dev-opt-im Cargo` profile; the full scalar matrix uses `--release`. Gates include formatting, all-target workspace `check/build` and `Clippy`; separate shared-CFF/Spenso feature configurations have their own receipts. UFO parity selects the API library with `ufo_support`; this is package-scoped optional-feature coverage. Cargo, nextest and Rayon use four workers under the recorded 30 GB process-tree guard. Final runtime selections allow up to ten seconds per process query; the guard still stops the pipeline on monitoring failure. Snapshot updates and automatic test retries are disabled. The full scalar matrix selects its

underlying namespaces directly, including slow cases. Exact arguments, warning policy, overrides and integrity guards remain authoritative in the ledger.

Dependency: Symbolica 2.2.0, with graphica and numerica, from `alpha00p/symbolica` at `4d0a833eb8e059d1f95bdae5abed2559830b235f`. This is a specific locked snapshot. Tool versions: `rustc` 1.97.0 (2d8144b78 2026-07-07); `cargo` 1.97.0 (c980f4866 2026-06-30); `cargo-nextest` 0.9.140. License material is excluded from reports and archives.

Source preservation and review scope

The mapping retains 37 remote and five review source attributions. Incoming `a1140c90c334ff58a3b040ff3eae06d3645bf0bb` is reviewed against `78395e3ab3ddd8d8f62b2f674d7488484eace197`: 68 paths (65 text, 3 binary), 792 text hunks, with explicit reviewer attribution. The three serialized Symbolica blobs received byte/hash and source-provenance review; their contents were not semantically decoded. At the tested candidate, exact correction fingerprints cover 21 changed files and the whole-stack inventory contains 647 commit/path records over 549 paths; inventory does not establish an unperformed fresh rereview of unchanged hunks.

Immutable corrected source `392c98c3d2f70896ef94267addfcf610e2df2c88` and reconciled C8 `449a59f6f173d100c2da858c541faa1e0dcca6e` share tree `c87cc95044663a68413010aceb81f9475be342c1` exactly. Later final-report changes require separate exact, reviewed documentation deltas from both the reconciled C8 and the executed candidate. Production/build/test/config content must remain identical. Candidate receipt hashes and execution counts are preserved; documentation closure does not count as another test run.

Limits and evidence

GL00/GL01 automatic route selections do not certify GL262. The source-scoped GL262 diagnostic reaches a Symbolica evaluator-construction panic with compilation disabled and supplies no complete matched runtime/performance result. No confirmed standalone reproducer of that large panic is established. The nonsymmetric `f(x, y)` import MRE demonstrates a separate argument-order defect and does not identify the panic's cause.

Fresh standalone evidence at executed revision `a3af927afc633c6e6c2474d3f03fa02e0242eaa4` has locked all-target check/build/Clippy passes, **two positive controls (39 and 4)**, and **one reproduced import defect**: the writer exits 0 and the reader exits the expected 101 for `f(y, x)` versus `f(x, y)`. These are not workspace test counts or a GL262 replay. Source-equivalent standalone package; no additional execution. The [standalone receipt](#) records exact package/manifest/lock/config identity and original binary/result hashes; carrying that evidence adds no execution. Separately recorded formatting passes at `bbb8ff1db499a78b3b25c41f774c9a034acb5b4d`; its tracked package/build inputs match observed C7.

Manual physical PySecDec integrations remain outside automatic coverage. Installed-Python subprocess tests and other platforms have no separate acceptance unless an explicit receipt records them. Finite coordinate probes and Monte Carlo acceptances have numerical/statistical limits; local Ward checks do not establish complete electroweak LU acceptance.

The [motivation audit](#) and [local UV measurements](#) retain workload/source/binary identities. On source `911f1824f994611d723dff0f50416cb301718617`, the local-only GL00/GL01 generation premiums are 4.45%/0.82%. A separate single-run scalar cohort includes integrated and threshold subtraction: GL21/base setup/generation takes 1.431860 times erased 3D. Other measured comparisons also show regressions; no general speedup is established. Those source-scoped observations do not certify this reconstructed stack. Stage timings, complete generation and saved-state sampling remain distinct.

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